Miles Reid NOTE

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1. MOTIVATION AND PRELIM THAT NOT MENTIONED IN THE BOOK

This one is reference to Miles Reid Notes, for me to easily read and memorized in my mind. I am very good at remembering but terribly slow to understand, so I need this note. This one also summarization to Week 1 until Week 5 of my lecturer's consultation.

1.1 FIELD EXTENSION

1.2 ALGEBRAIC FIELD

Using

1.3 SOME DISCUSSED THEOREM THAT WILL NOT BE MENTIONED

Here is kinda important lemma, sometimes treated as definition in some book

$$\mathfrak{m} \text{ is maximal ideal of ring K } \Leftrightarrow K/\mathfrak{m} \text{ is a field}$$

2. AFFINE VARIETIES

2.1 NOETHERIAN RINGS

Definition 2.1 (Noetherian)**.** A ring R is called Noetherian if every ideal of R is finitely generated.

Proposition 2.2. *Noetherian rings has equivalent statements*

- *Every ideal of R is finitely generated, i.e. for an ideal I of R, there exists $f_1, f_2, ..., f_n \in R$ such that*

$$I = (f_1, f_2, \ldots, f_n)$$

- *For every ascending chain*

$$I_1 \subset I_2 \subset \ldots \subset I_m \subset \ldots$$

  of ideal ring R terminates, i.e. there exist I_N such that $I_N = I_{N+1}$

- *Every non-empty set of ideal of R has maximal element.*

Theorem 2.3 (Hilbert Basis Theorem)**.** *Let R be a Noetherian ring. Then the polynomial ring $R[x]$ is also Noetherian.*

3. Zariski Topology

3.1 Algebraic Set

3.2 The Corresponding of Algebraic Set

3.3 Irreducible Algebraic Set

4. Nullstellensatz

4.1 Radical Ideals

4.2 Nullstellensatz theorems

Or also know as Hilbert's zeros theorem.

Theorem 4.1 (Weak Nullstellensatz, algebraic form)**.** *Suppose k is an algebraically closed field, and let $\mathfrak{m}$ is maximal ideal of $k[x_1, \ldots, x_n]$, then we have*

$$\mathfrak{m} = (x_1 - a_1, \ldots, x_n - a_n)$$

Proof. Let $\mathfrak{m}$ be maximal ideal of $k[x_1, \ldots, x_n]$, then we have $K = k[x_1, \ldots, x_n]/\mathfrak{m}$ is a field. Consider

$$\begin{array}{ccc} k & & \\ & \searrow^{\varphi} & \\ k[x_1, \ldots, x_n] & \longrightarrow & K \end{array}$$

Notice that ψ is field extension and since k is algebraically closed, then ψ is iso. □

Theorem 4.2 (Weak Nullstellensatz, geometric form)**.** *Suppose k is an algebraically closed field. Let I be a proper ideal in the polynomial ring $k[x_1, \ldots, x_n]$. Then we have*

$$V(I) \neq \emptyset$$

i.e., there exists a point $(a_1, \ldots, a_n) \in k^n$ such that $f(a_1, \ldots, a_n) = 0$ for all $f \in I$.

Theorem 4.3 (Strong Nullstellensatz)**.** *Suppose k is an algebraically closed field. Let I be an ideal in the polynomial ring $k[x_1, \ldots, x_n]$. Then the ideal of all polynomials that vanish on the variety defined by I is equal to the radical of I, i.e.,*

$$I(V(I)) = \sqrt{I}.$$

4.3 COROLLARY OF NULLSTELLENSATZ